Progress Report — STF AI Vehicle Diagnosis Platform

Reporting period: 6 June – 27 June 2026 (last three weeks) | **Prepared for:** Project Management | **Author:** Xiangzhu Yan

1. Executive Summary

Over the last three weeks the team moved the AI diagnosis product from “it runs” to “it runs reliably and we can measure how good it is.” Three storylines dominated the period:

1. **The product now shows its work in real time.** Users previously stared at a frozen spinner for minutes while the AI thought. It now streams its reasoning live, so the experience feels responsive and trustworthy.
2. **We fixed the accuracy root-cause: the AI didn’t know which vehicle it was looking at.** We made vehicle make/model mandatory on every upload and forced the AI to ground its answers in the correct service manual — directly attacking an earlier failure where the system mistook one vehicle for a completely different one.
3. **We built our first scorecard.** For the first time we can put a number on diagnosis quality and compare our new “AI agent” approach against the older method. The agent won decisively.

> **By the numbers:** ~20 changes shipped and merged • 10 tracked tickets closed • ~15,500 lines of code added across backend, evaluation harness, and documentation • zero changes left unmerged or stuck.

2. What We Delivered (grouped by business value)

A. A more trustworthy, responsive user experience

- **Live progress streaming** (HARNESS-22) — the AI now narrates its reasoning step-by-step instead of showing a silent spinner. This was the single biggest perceived-quality complaint, and it’s resolved.
- **First-diagnosis-after-deploy no longer fails** (APP-58) — previously the first diagnosis after a server update died with a “network error” because the AI model took minutes to wake up. We added an automatic warm-up and a keep-alive signal so the first real user request always succeeds.

B. Fixing diagnosis accuracy at the root

- **Mandatory vehicle identification** (APP-59, APP-60) — every uploaded data file and manual must now declare its manufacturer and model. The AI can no longer guess.
- **“Honest” AI that stays grounded** (HARNESS-25, HARNESS-26) — the AI is constrained to reason from the correct vehicle’s manual, and to admit when it lacks the information rather than inventing an answer.
- **Factory-code matching** (APP-61) — real manuals use internal factory codes (e.g. the “Tricity 155” is internally “MWS150-A”). We taught the system these aliases so it finds the right manual even when the names don’t match on the surface.

C. Measuring quality for the first time (our new scorecard)

- **First baseline evaluation** (HARNESS-23) — we assembled 30 locked “gold-standard” diagnostic questions and scored two approaches against them. **Result: the new AI-agent approach scored 0.59 vs. 0.34 for the older retrieval method — a ~75% relative improvement, winning on every question type.** This gives us a defensible number to track from now on.

- **Identified the agent’s main weakness** — the agent sometimes runs out of its time/step budget before finishing. We’ve started addressing this (more room to work, and teaching it to stop and conclude gracefully when information is genuinely absent).

D. Reliability & housekeeping

- **Data quality fix** (APP-57) — records with corrupt timestamps are now dropped cleanly instead of being silently mislabeled with the current time.
- **Feedback on AI diagnoses now works** (HARNESS-24) — fixed a bug where users couldn’t submit feedback on agent diagnoses.
- **Removed dead/legacy code** (APP-53) and published a verified architecture overview so new contributors and reviewers can understand the system quickly.

3. In Progress / Not Yet Done

- **Diagnosis quality is still being raised.** The 0.59 baseline is a starting point, not a finish line. A backlog of ~19 follow-up improvements is queued, with a re-measurement planned as the gate before declaring the next milestone.
- **A known speed bottleneck remains.** The local AI model has a “thinking” mode that is hard to switch off and slows some evaluation runs. We’ve worked around it in one path; a fuller fix is still open.
- **Corpus coverage gap.** Our reference-manual library is still narrow (a couple of vehicle models). The AI is only as good as the manuals it can read — expanding this library is a key dependency for broader accuracy.

4. Risks & Watch Items

Item	Why it matters	Status
Manual / corpus coverage	AI can’t diagnose vehicles it has no manual for	Open dependency
Local model speed	Slows evaluation cycles; affects responsiveness under load	Partially mitigated
Quality still below target	0.59 baseline needs to climb before pilot-ready	Active backlog

5. Bottom Line

The platform graduated from a fragile demo to a measurably-improving product this period. We can now (a) show users the AI is working, (b) trust it to identify the right vehicle, and (c) **prove** quality with a repeatable score. The next phase is straightforward: push that score up and broaden the manual library.